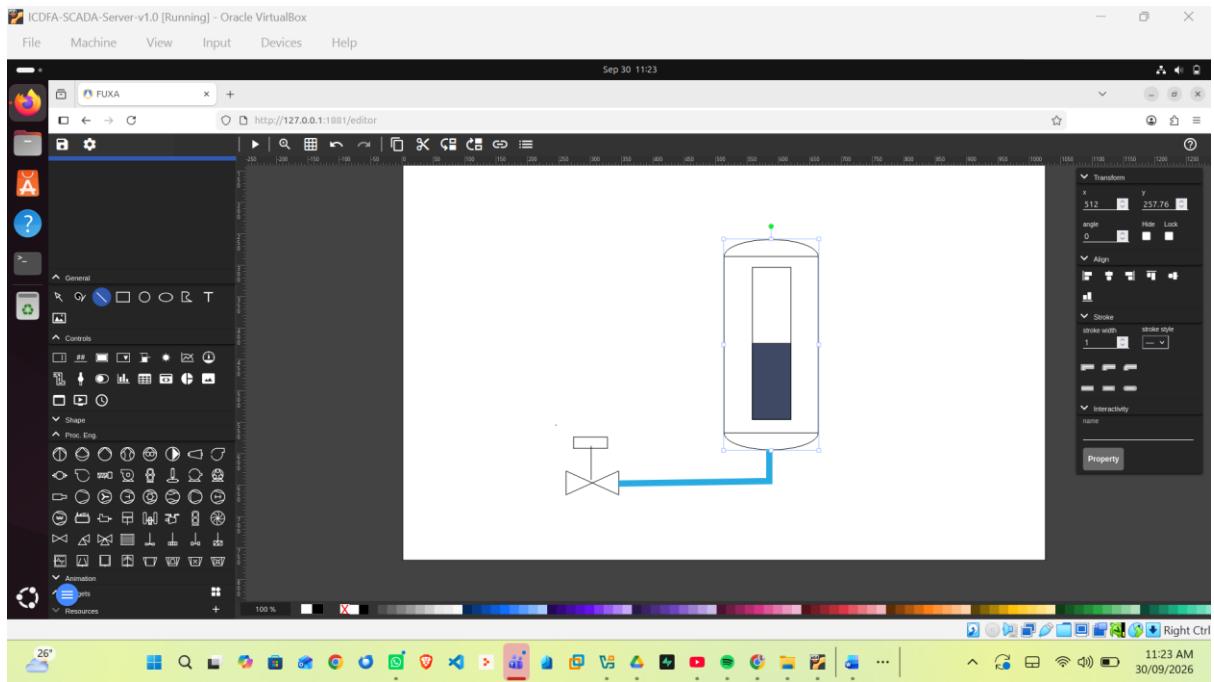
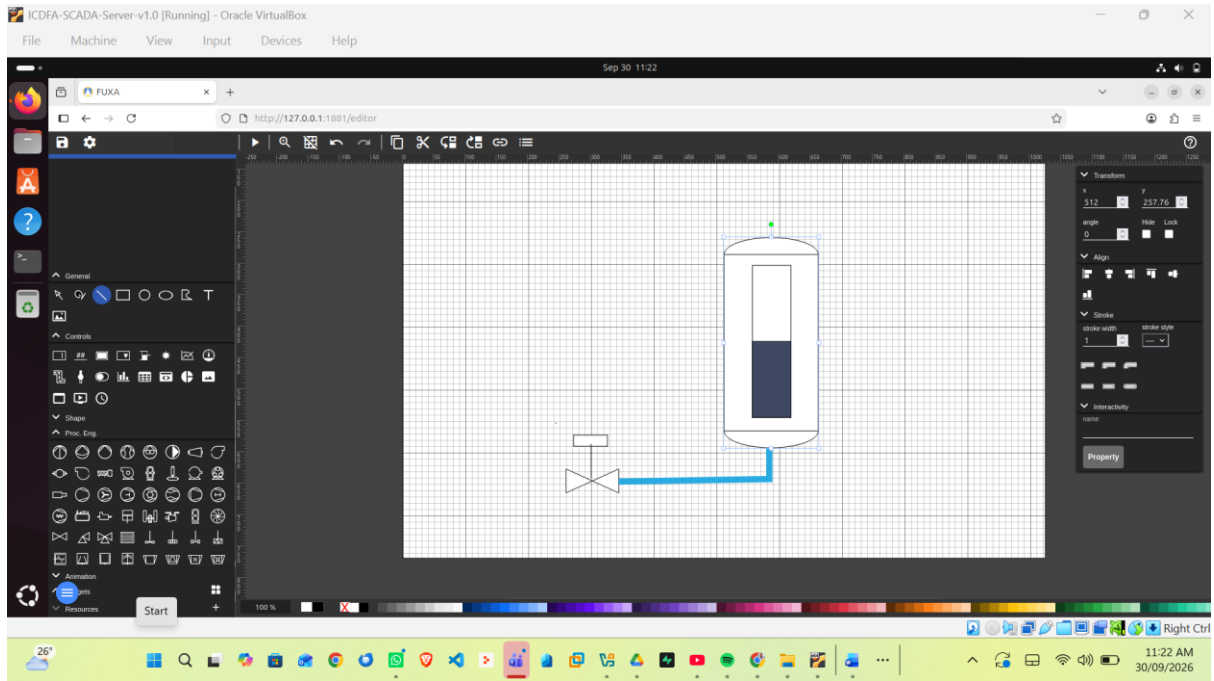
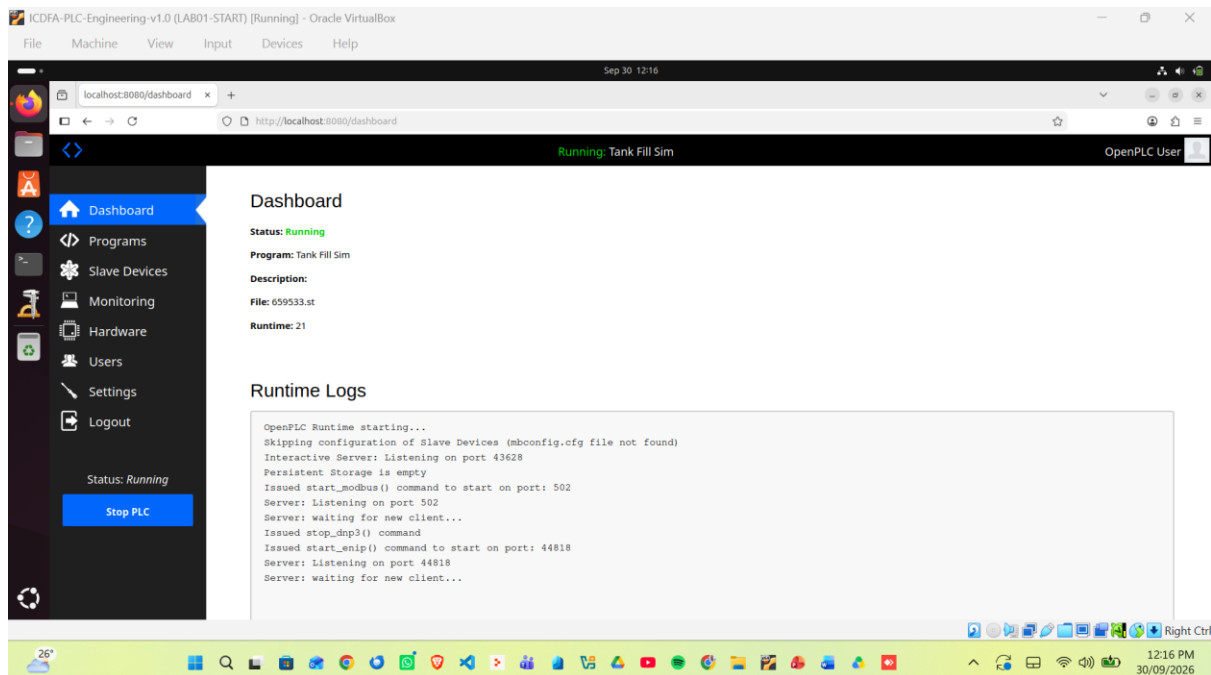




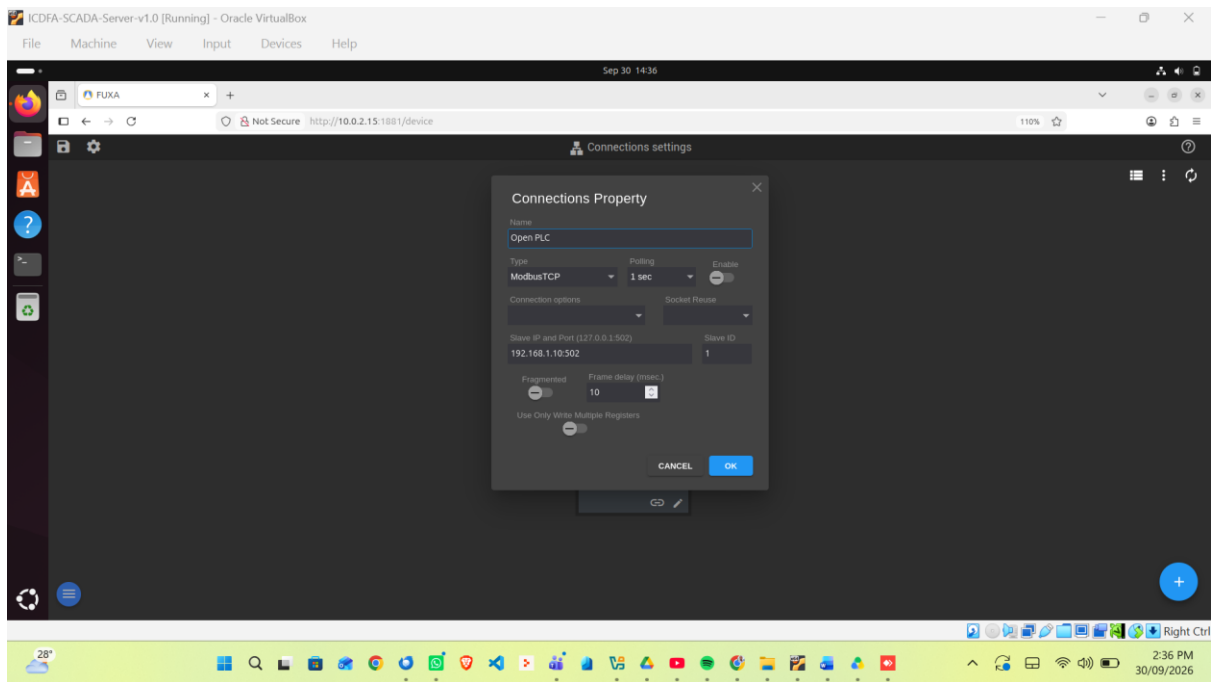
HMI Design/Edit Mode

This screenshot shows the HMI design environment in FUXA edit mode with the tank filling process interface being developed. The grid layout is enabled to assist with accurate positioning and alignment of graphical components. The interface contains the main process elements, including the storage tank, valve, and connecting pipe, which represent the physical process layout. This stage demonstrates the creation of the HMI visualization based on industrial process representation principles.



OpenPLC Tank Fill Simulation Running

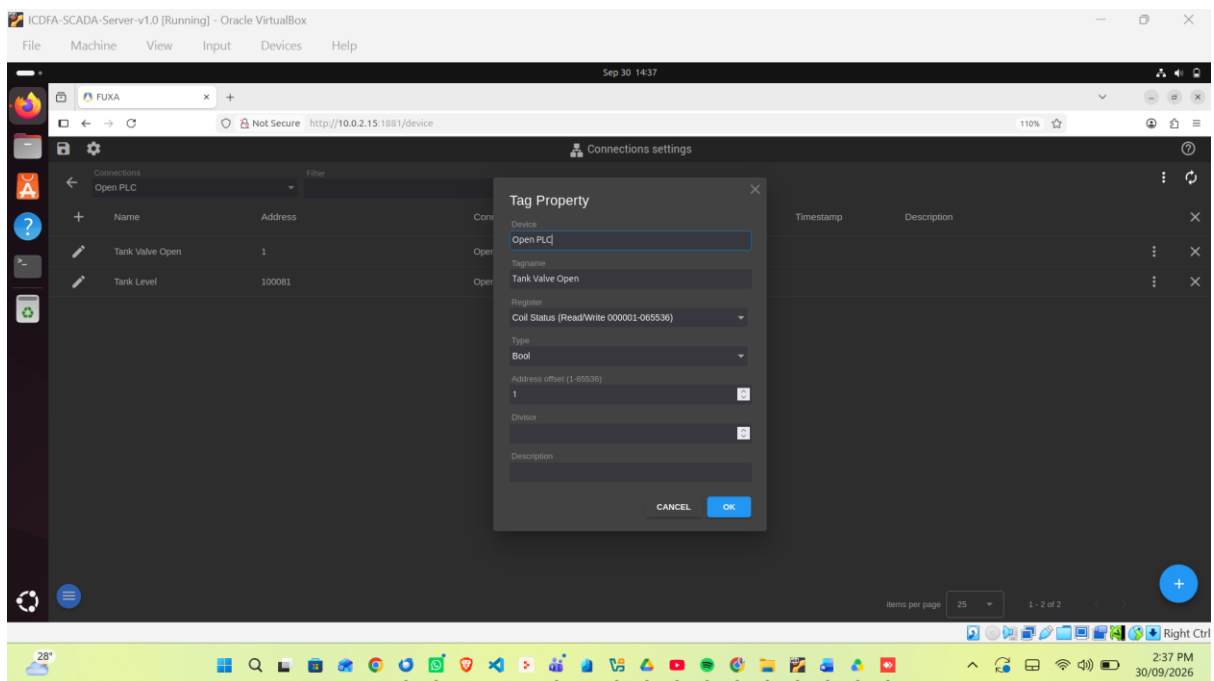
This screenshot shows the OpenPLC runtime environment with the Tank Fill Simulation program successfully running. The dashboard confirms that the PLC status is active and the uploaded Tank Fill Sim program is executing. This step verifies that the PLC is ready to communicate with the HMI and provide real-time process data for visualization and control.



PLC Device Connection Configuration

This screenshot shows the configuration of the OpenPLC device connection in the FUXA SCADA environment. The HMI was configured to communicate with the PLC using the Modbus TCP protocol. The connection parameters, including the PLC address and communication settings, were defined to establish data exchange between the OpenPLC runtime and the HMI system.

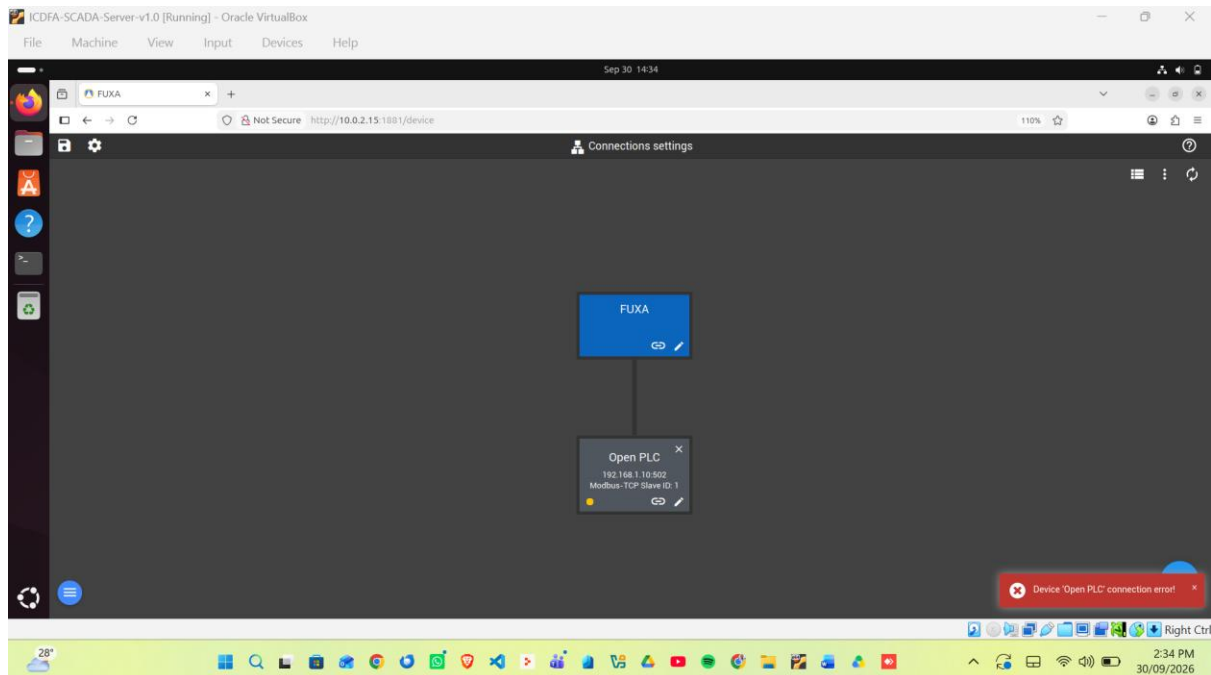
This connection allows FUXA to retrieve real-time process information from the PLC and use the data for HMI visualization and control.



Configured PLC Tags

This screenshot shows the configured PLC tags within the FUXA HMI system. The Tank Valve Open tag has been added and linked to the OpenPLC device. The tag configuration defines the data source and data type required for communication with the PLC.

The Tank Valve Open tag is used to monitor and animate the valve state in the HMI. A Boolean data type is used because the valve has two operating conditions: open and closed.



Knowledge Check

1. What is a P&ID and how does it influence HMI layout?

A P&ID (Process and Instrumentation Diagram) is an engineering drawing that represents an industrial process by showing process equipment, piping, instruments, control devices, and their relationships.

A P&ID influences HMI layout because the HMI is often designed based on the process arrangement shown in the P&ID. The graphical elements on the HMI, such as tanks, valves, pipes, and measurement instruments, are arranged to represent the real industrial process, making it easier for operators to understand and monitor the system.

2. Why do high-performance HMIs avoid unnecessary animation and excessive color?

High-performance HMIs avoid unnecessary animation and excessive colors because they can distract operators and make important information difficult to identify.

A simple design using neutral backgrounds and limited colors allows abnormal conditions, alarms, or process problems to stand out clearly. This helps operators quickly focus on areas requiring attention and improves decision-making.

3. How does the Tank Level tag differ from the Tank Valve Open tag in data type and behavior?

The Tank Level tag represents the measured level of liquid inside the tank. It is an analog value because the level can continuously change within a range (for example, from empty to full).

The Tank Valve Open tag represents the operating state of the valve. It is a Boolean/digital value because it has only two possible states:

- 0 = Valve closed
- 1 = Valve open

The Tank Level changes gradually based on the filling process, while the Tank Valve Open tag switches between two fixed conditions.

4. What could happen operationally if an HMI displays stale or incorrect PLC data?

If an HMI displays stale or incorrect PLC data, operators may make decisions based on inaccurate information about the actual process condition.

Possible consequences include:

- Incorrect control actions
- Failure to detect faults or abnormal conditions
- Delayed response to process problems
- Equipment damage
- Reduced operational safety

Accurate and real-time communication between the PLC and HMI is therefore important for effective monitoring and control of industrial processes.